SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:**CSA1515

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct

I Sanjay S/192521058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: Sanjay S

Assessment Tasks
Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
   The system must handle millions of concurrent users with minimal buffering.
   Design a cloud architecture using scalability and caching techniques.
   Explain how load balancing and auto-scaling can ensure smooth streaming.
   Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
   They face challenges in maintaining consistency and managing deployments.
   Propose a solution using container orchestration and multi-cloud strategies.
   Explain how service discovery and scaling can be handled efficiently.
   Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
   The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning.
Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
Some applications must remain on-premises due to compliance requirements.
Design a hybrid architecture that ensures secure communication between environments.
Explain how identity management and encryption should be implemented.
Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
The system must continue operating even if one region goes down.
Propose a multi-region deployment strategy in the cloud.
Explain how data replication and failover mechanisms should be configured.
Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1. Cloud Architecture for a Video Streaming Platform

1. Understanding of Scenario

A media company needs to support millions of concurrent users with minimal buffering. The system must scale automatically during peak demand while maintaining video quality and controlling cloud costs.

2. Identification of Risks / Issues

- High traffic may overload application servers.
- Network congestion and distant video sources may cause buffering.
- Sudden traffic spikes may exceed available computing capacity.
- Excessive bandwidth, storage, and compute usage may increase costs.

3. Application of Concepts

Millions of Concurrent Users
Global DNS / Traffic Routing
CDN Edge Caching
Load Balancer
Auto-Scaling Servers
Video Storage and Transcoding

- Scalability: Use horizontal scaling to add application instances when demand increases.
- Caching: Store popular video segments at CDN edge locations close to users, reducing latency and origin-server load.
- Load balancing: Distribute requests across healthy application servers.
- Adaptive bitrate streaming: Use HLS or MPEG-DASH so video quality adjusts to the user's available bandwidth.
- Auto-scaling: Scale instances based on request rate, CPU usage, queue length, and latency.

4. Decision Justification

Use a CDN-first architecture with object storage, adaptive bitrate streaming, load balancing, and automatic scaling. Cache popular content at the edge and use lifecycle policies to move older content to cheaper storage. Apply reserved capacity or savings plans for predictable workloads and on-demand or spot capacity for suitable non-critical tasks, such as some transcoding jobs.

5. Clarity

Conclusion: A CDN combined with auto-scaling and load balancing minimizes buffering by reducing latency and distributing traffic. Cost optimization should focus on bandwidth, caching efficiency, storage tiers, and demand-based resource allocation.

2. Multi-Cloud Container Deployment for a Fintech Startup

1. Understanding of Scenario

A fintech startup deploys containerized applications across multiple cloud providers. It needs consistent environments, reliable deployment, efficient scaling, and secure communication between services.

2. Identification of Risks / Issues

- Differences in cloud configurations may cause inconsistent deployments.
- Service discovery and networking may be difficult across providers.
- Financial data requires strong security and regulatory compliance.
- Multi-cloud monitoring and management can increase complexity and cost.

3. Application of Concepts

CI/CD Pipeline + Container Registry
Cloud A
Kubernetes cluster
Cloud B
Kubernetes cluster
Centralized Monitoring • Secrets • Policy Management

- Container orchestration: Use Kubernetes to manage container scheduling, health checks, rolling updates, and recovery.

- Consistency: Build immutable container images and deploy the same tested image digest across environments using infrastructure as code.
- Service discovery: Use Kubernetes Services and DNS within each cluster; use a service mesh or controlled cross-cloud gateway where cross-cloud communication is required.
- Scaling: Configure Horizontal Pod Autoscaler and cluster autoscaling based on CPU, memory, request rate, and queue depth.
- Security: Use workload identities, secret management, mutual TLS where appropriate, and least-privilege access.

4. Decision Justification

Adopt a Kubernetes-based deployment model with a shared CI/CD pipeline and infrastructure-as-code templates. Keep cloud-specific configurations in controlled environment overlays. Use multi-cloud deployment for disaster recovery or selected critical services rather than duplicating every workload without a clear need.

Benefits: Portability, reduced dependence on one provider, improved resilience, and flexible placement of workloads.

Risks: Higher operating costs, network latency, duplicated security controls, inconsistent services, and more complex troubleshooting.

5. Clarity

Conclusion: Standardized container images, Kubernetes orchestration, automated pipelines, and consistent security policies improve deployment reliability. Multi-cloud is beneficial when its resilience and portability justify the extra complexity and cost.

2. Backup and Disaster Recovery for an E-Learning Platform

1. Understanding of Scenario

An e-learning platform loses data because of accidental deletion. It needs to restore courses, student records, assignments, and other critical information while minimizing downtime and data loss.

2. Identification of Risks / Issues

- Accidental deletion or incorrect updates.
- Ransomware or unauthorized modification.
- Regional outages or storage failures.
- Backups that are corrupted, outdated, or inaccessible during recovery.

3. Application of Concepts

Primary Cloud Database and Object Storage
Versioning and Point-in-Time Recovery
Automated Backups
Separate Account / Region with Immutable Backup Copies
Restore, Validate, and Resume Service

- Enable object versioning and database point-in-time recovery.
- Schedule automated backups according to data criticality.
- Replicate backup copies to a separate region or account.
- Use immutable or write-once backup retention to resist accidental deletion and ransomware.
- Regularly test restoration and verify data integrity.

RTO (Recovery Time Objective): The maximum acceptable time to restore service after an outage.
RPO (Recovery Point Objective): The maximum acceptable amount of data loss, expressed as time.
For example, an RTO of 1 hour and an RPO of 15 minutes mean the platform aims to restore service within 1 hour and lose no more than 15 minutes of data. These are illustrative targets and must be validated through recovery tests.

4. Decision Justification

Use a combination of automated backups, point-in-time recovery, versioning, and cross-region immutable copies. Replication alone is not enough because accidental deletion can replicate to the secondary system too. A recovery runbook should cover failover, restoration, integrity checks, communication, and service validation.

5. Clarity

Conclusion: A layered backup and disaster recovery strategy protects against deletion, corruption, and regional failure. Measured recovery tests ensure that the defined RTO and RPO can actually be achieved.

4. Hybrid Cloud Architecture for a Government Agency

1. Understanding of Scenario

A government agency needs to store sensitive information on-premises because of compliance requirements while using public cloud services for non-sensitive applications. Secure communication and consistent access control are essential.

2. Identification of Risks / Issues

- Unauthorized access to sensitive government information.
- Data leakage during transfers between environments.
- Misconfigured access policies or encryption.
- Network interruptions and difficulties maintaining consistent security across environments.

3. Application of Concepts

On-Premises
Classified or restricted data
Public Cloud
Approved non-sensitive services
Private Connectivity / VPN + Firewalls + API Gateway
Central Identity Governance • Encryption • Audit Logging

- Secure communication: Use private connectivity or IPsec VPN, firewalls, network segmentation, and authenticated APIs.
- Identity management: Integrate a trusted identity provider, enforce MFA, and apply role-based access control and least privilege.
- Encryption: Use TLS for data in transit and strong encryption for stored data. Protect encryption keys using controlled key-management systems.
- Data governance: Classify information and prevent sensitive data from entering public cloud services unless explicitly permitted.
- Monitoring: Collect audit logs from both environments and monitor suspicious access.

4. Decision Justification

Use a hybrid model in which restricted applications and data remain on-premises, while approved public cloud workloads use scalable cloud resources. Establish common identity policies, encryption standards, centralized logging, and documented compliance controls.

Challenges: Integration complexity, inconsistent configurations, network latency, skills requirements, and duplicated operational responsibilities.

5. Clarity

Conclusion: Hybrid cloud balances compliance and scalability when data classification, secure connectivity, centralized identity governance, and encryption are consistently enforced across both environments.

5. Multi-Region High Availability for a SaaS Application

1. Understanding of Scenario

A SaaS provider must keep its application available even if one cloud region fails. It needs redundancy, reliable data replication, automated failover, and proactive monitoring.

2. Identification of Risks / Issues

- A regional outage may make the application unavailable.
- Replication delays may cause data loss or stale reads.
- Incorrect failover configuration may route users to an unhealthy region.
- Split-brain writes or configuration inconsistencies may corrupt data.
- Poor monitoring may delay detection and recovery.

3. Application of Concepts

Global Traffic Manager / Health-Based Routing
Region A
Application + Database
Primary
Region B
Application + Replicated Data
Standby or active
Replication • Health Checks • Automated Failover

- Multi-region deployment: Deploy application instances in at least two regions, preferably across

independent failure domains.

- Data replication: Use synchronous replication where supported and appropriate for strict consistency, or asynchronous replication when latency and distance make it preferable.
- Failover: Route traffic to healthy regions using health checks and automated traffic management.
- Database protection: Use a clear primary-writer or consensus-based design to prevent conflicting writes. Define promotion and failback procedures.
- Monitoring: Track availability, latency, error rates, replication lag, saturation, and failed health checks.
- Alerting: Send actionable alerts to the operations team and maintain tested incident-response runbooks.

4. Decision Justification

For most SaaS platforms, a multi-region active-passive design is a practical starting point: the secondary region is ready to take over when the primary fails. Active-active can improve global latency and utilization but requires more complex data consistency and conflict handling.

Regularly test regional failover, backups, DNS or traffic-routing changes, and recovery procedures. Define RTO and RPO targets based on business requirements.

5. Clarity

Conclusion: Multi-region deployment improves availability by removing dependence on a single region. Replication, health-based routing, tested failover, and proactive monitoring help maintain service continuity and reduce recovery time.